# Deploying Web App on ECS – Step 3: Create Amazon ECS Cluster

# To Begin with the Lab

## Summary of the Lab

In this step, a new **Amazon ECS** cluster was created using **AWS Fargate** as the infrastructure provider. Before creating the new cluster, any existing cluster was deleted so the complete cluster creation process could be demonstrated from scratch. The lab showed that an ECS cluster acts as the logical environment where ECS tasks and services are deployed, while **AWS Fargate** automatically manages the underlying compute infrastructure without requiring EC2 instances. After the cluster was created successfully, it became ready for deploying the application and configuring the **Application Load Balancer** in the next step.

## Understanding Amazon ECS Cluster

An **Amazon ECS Cluster** is a logical grouping of resources where **Amazon ECS** runs containerized applications as tasks and services. It exists to organize and manage workloads while deciding where containers should run. When **AWS Fargate** is used, AWS automatically provisions and manages the compute infrastructure, so there is no need to launch or maintain EC2 instances. This allows developers to focus on deploying applications instead of managing servers.

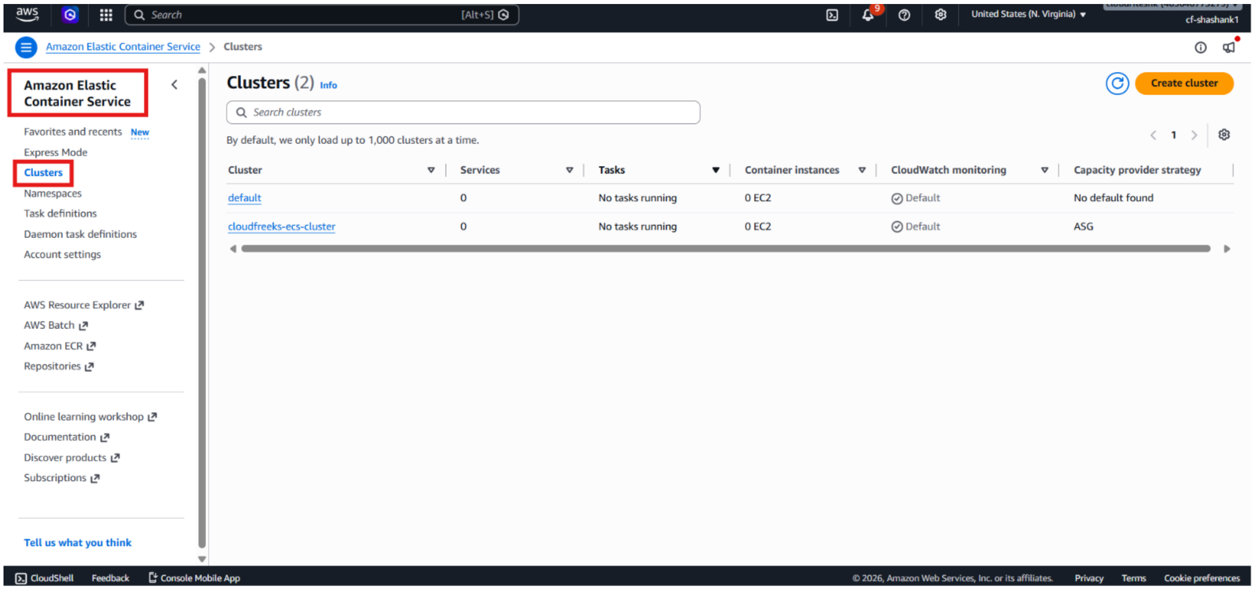
**Example:**

A company wants to deploy its PHP web application without maintaining virtual machines. Instead of creating EC2 instances, they create an **Amazon ECS** cluster using **AWS Fargate**. Later, when the application service is deployed, Fargate automatically provides the required compute resources and runs the containers inside the cluster.

## Lab Steps

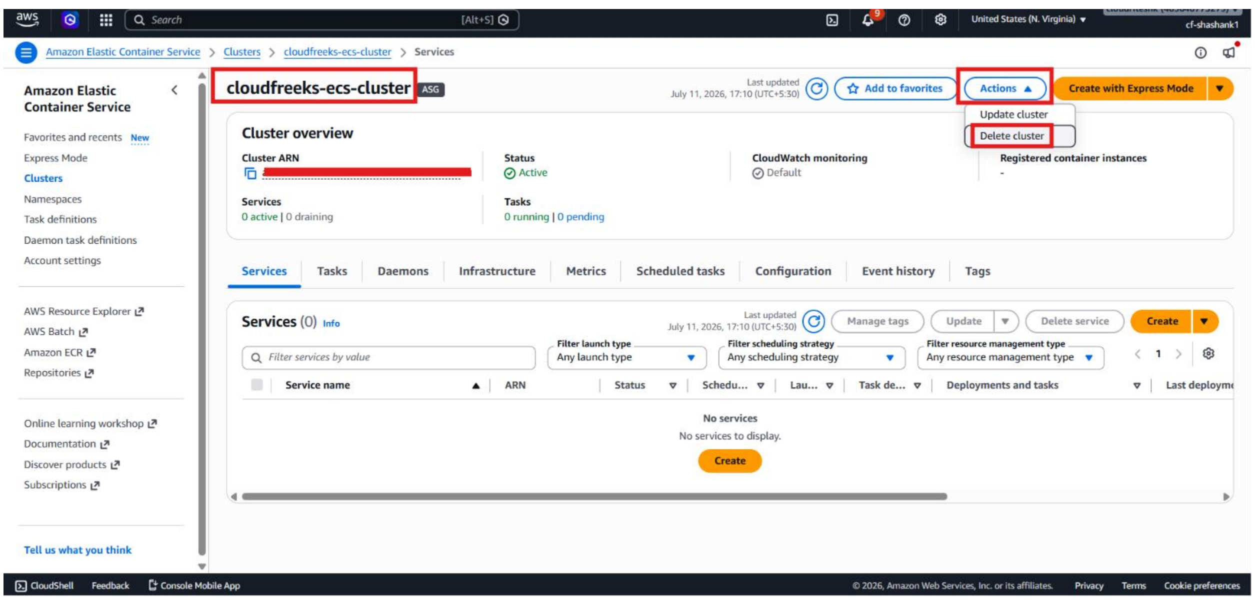
### Step 1 – Open Amazon ECS

- Sign in to the **AWS Management Console**.
- Navigate to **Amazon ECS**.
- Open the **Clusters** section.




## Step 2 – Remove the Existing Cluster (Optional)

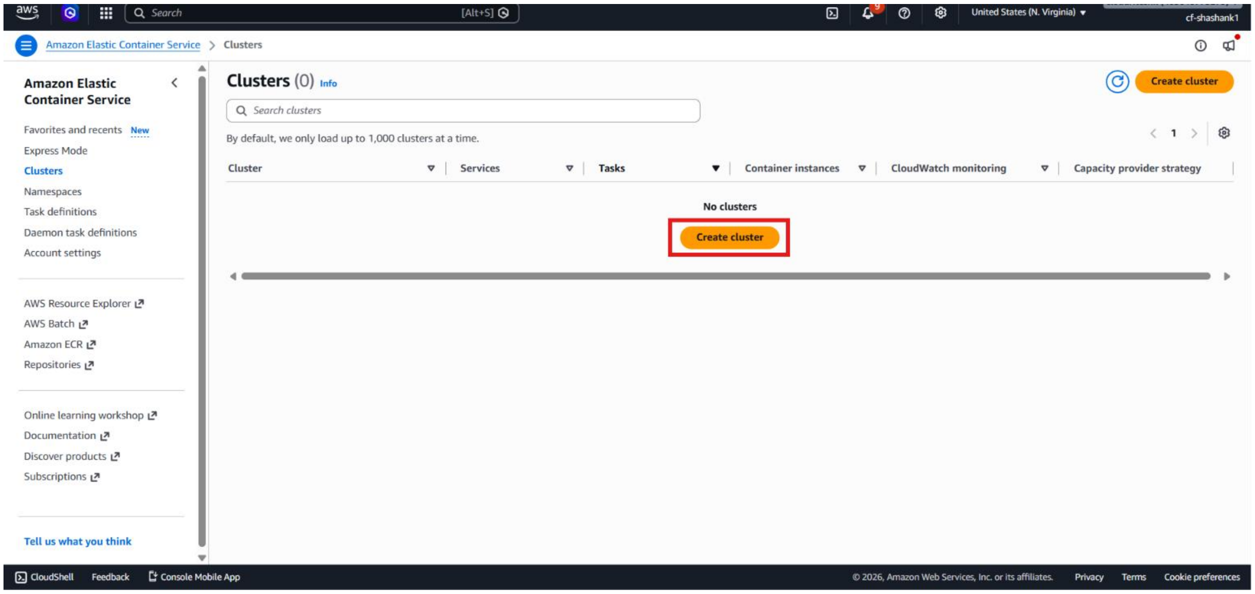
- Review the existing ECS clusters.
- Select the previously created cluster if it exists.
- Click **Delete cluster**.



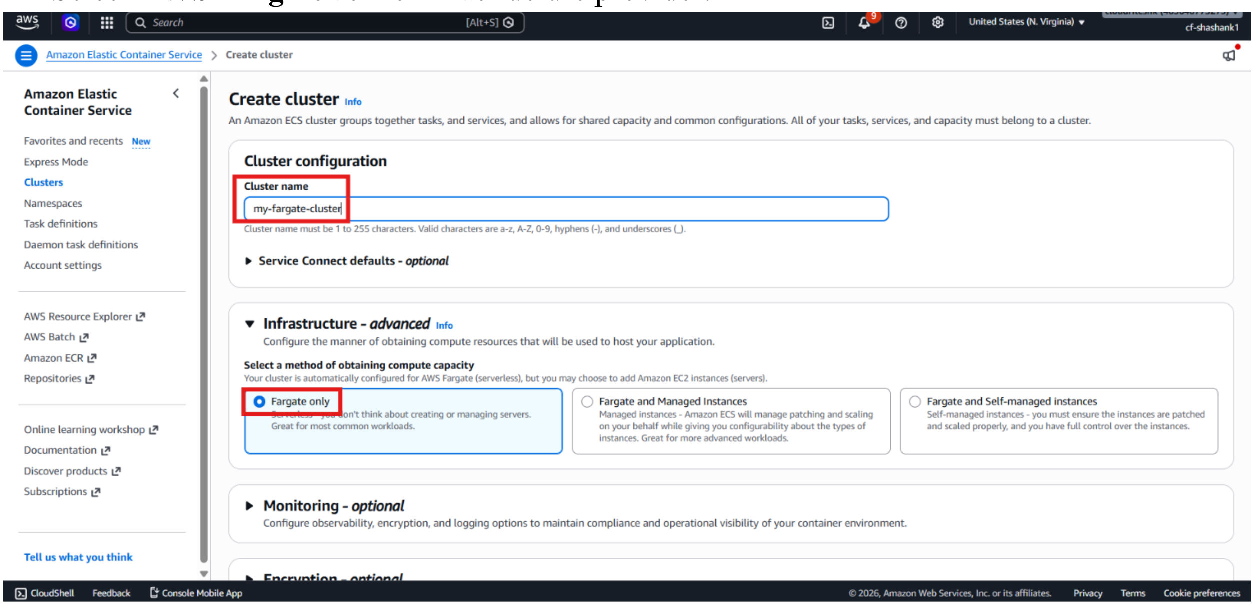

- Wait until the cluster is deleted successfully.
- Skip this step if no cluster already exists.

## Step 3 – Create a New ECS Cluster

- Click **Create cluster**.



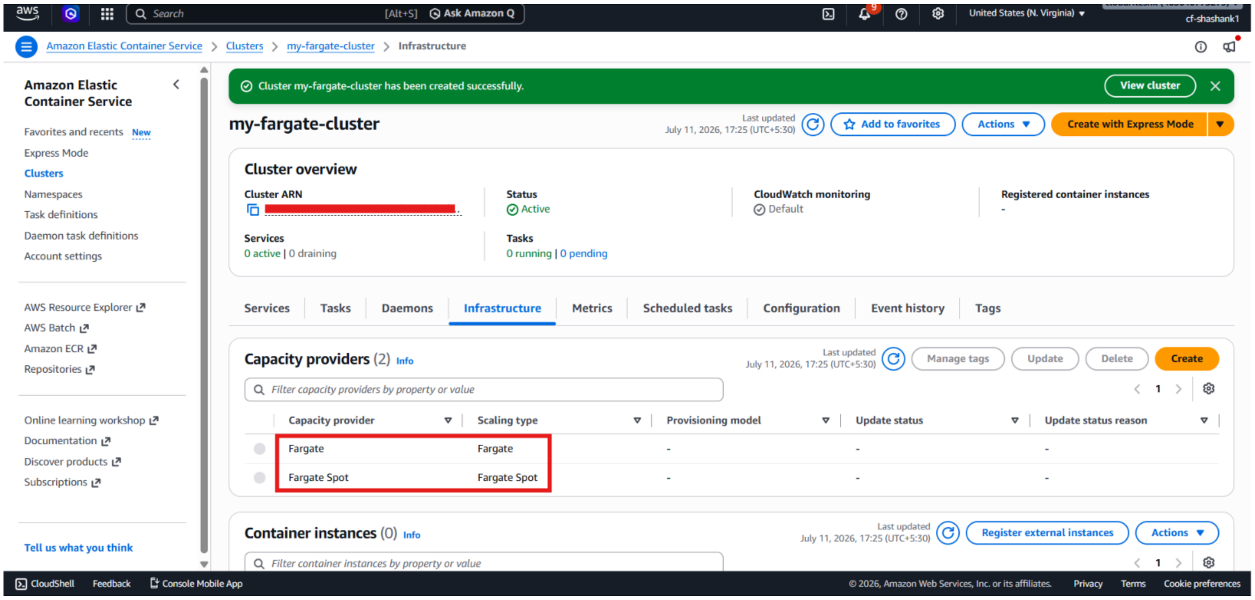

- Enter the cluster name: my-fargate-cluster
- Select **AWS Fargate** as the infrastructure provider.




- Leave the remaining configuration with the default values.
- Click **Create**.

## Step 4 – Verify the Cluster Configuration

- Wait until the cluster creation process completes.
- Open the newly created cluster.
- Verify that **AWS Fargate** is listed as the infrastructure.




- Confirm that no EC2 container instances are attached to the cluster.

## Verification

- Verified that the previous ECS cluster was deleted successfully (if one existed).
- Verified that a new **Amazon ECS** cluster was created successfully.
- Verified that **AWS Fargate** was selected as the infrastructure provider.
- Verified that no EC2 container instances were required or attached.
- Verified that the ECS cluster is ready for deploying ECS services and integrating with the **Application Load Balancer**.